

Simple document

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I'm an R Markdown document!

Section 0: libraries

```
## Warning: package 'tidyverse' was built under R version 4.4.3
```

Section 1

Here's a **code chunk** that samples from a *normal distribution*:

```
samp = rnorm(100)
length(samp)
```

```
## [1] 100
```

Section 2

I can take the mean of the sample, too! The mean is -0.0420658.

Section 3: a tibble

```
plot_df = tibble(
  x = rnorm(1000, sd = .5),
  y = 1 + 2 * x + rnorm(1000)
)

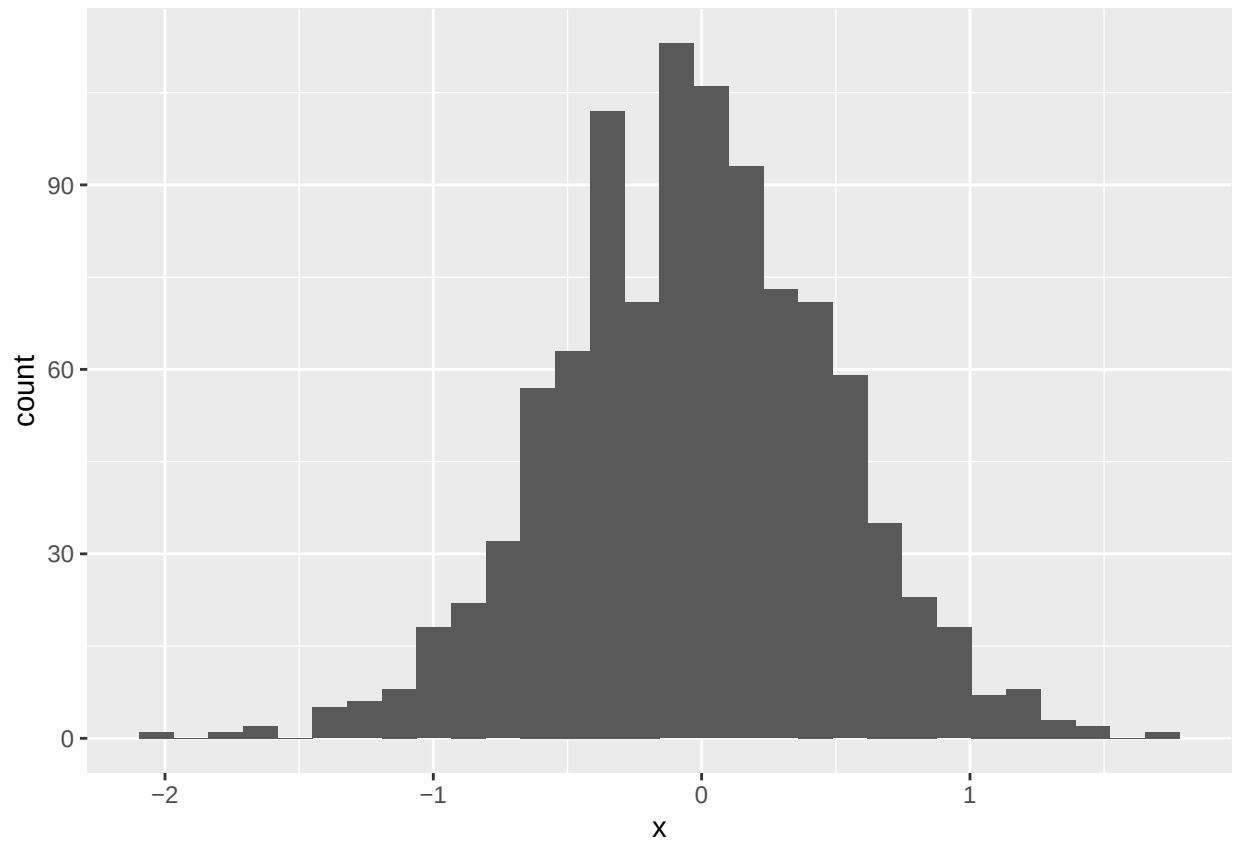
head(plot_df)
```

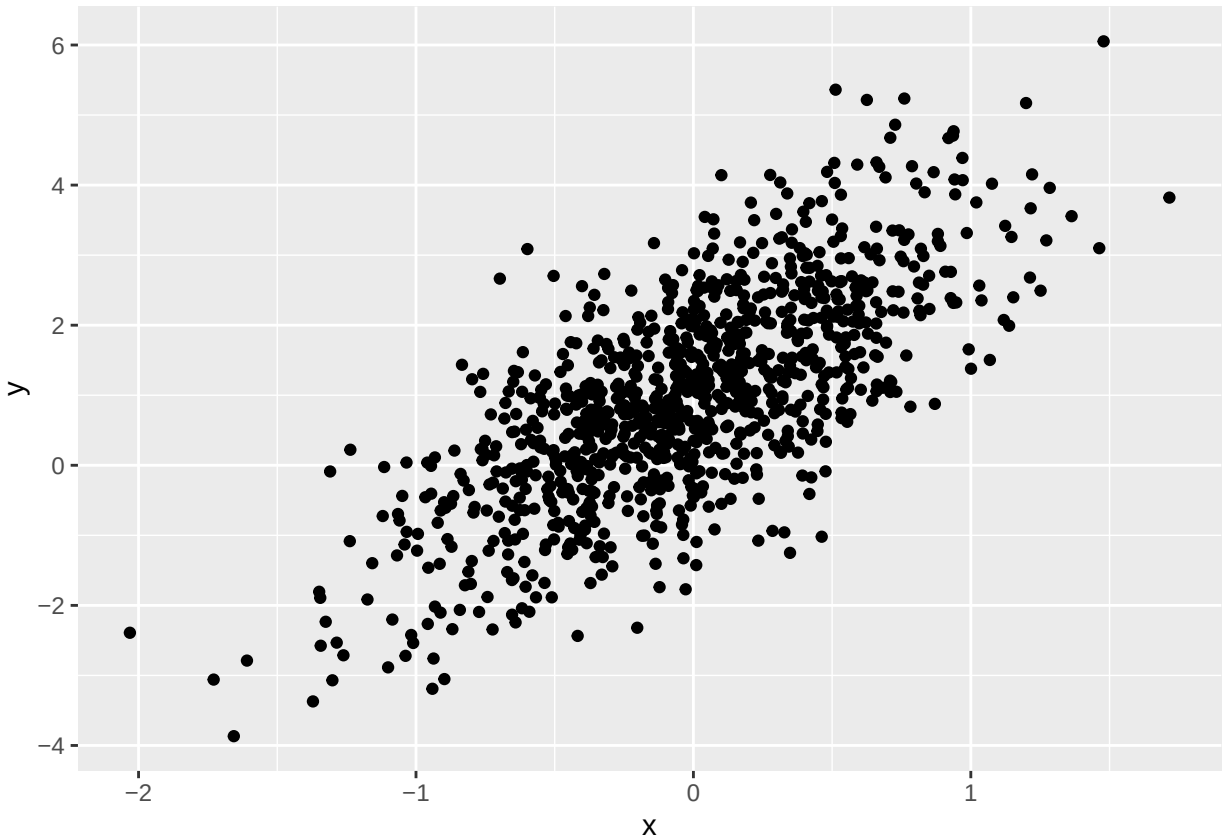
```
## # A tibble: 6 x 2
##       x       y
##   <dbl> <dbl>
## 1  0.920  4.67
## 2  1.02   3.75
## 3 -0.745 -0.646
## 4 -0.391  0.644
## 5  0.102  0.805
## 6 -0.0737 2.57
```

Section 4: plots

These are plots from are random sample.

```
## 'stat_bin()' using 'bins = 30'. Pick better value 'binwidth'.
```





Section 5: Learning Assessment 2

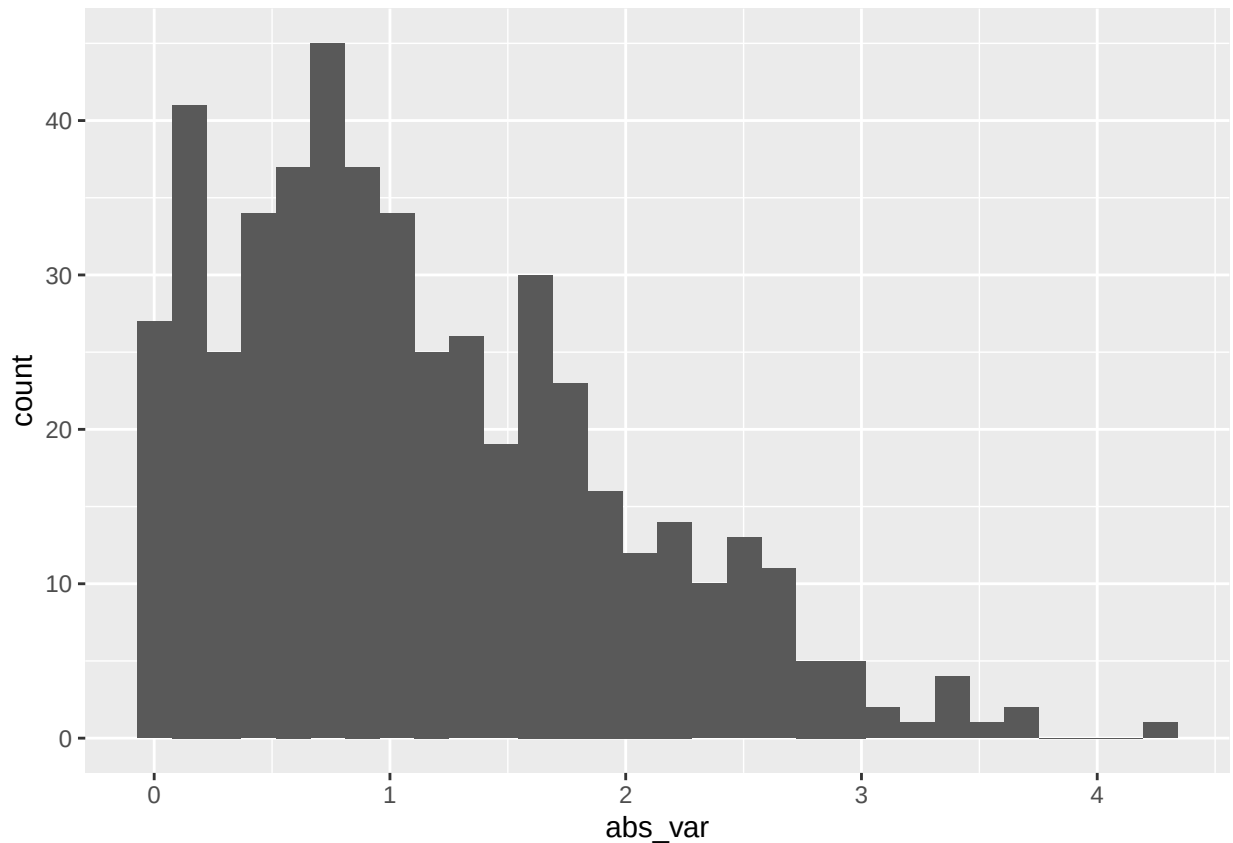
Write a named code chunk that creates a dataframe comprised of: a numeric variable containing a random sample of size 500 from a normal variable with mean 1; a logical vector indicating whether each sampled value is greater than zero; and a numeric vector containing the absolute value of each element. Then, produce a histogram of the absolute value variable just created. Add an inline summary giving the median value rounded to two decimal places. What happens if you set `eval = FALSE` to the code chunk? What about `echo = FALSE`?

This plot shows the distribution of the absolute value of $X \sim N(1, 1)$

```
set.seed(0)
assessment_df = tibble(
  num_var = rnorm(500, mean = 1),
  log_var = num_var > 0,
  abs_var = abs(num_var),
)

ggplot(assessment_df, aes(x = abs_var)) +
  geom_histogram()
```

```
## 'stat_bin()' using 'bins = 30'. Pick better value 'binwidth'.
```



```
median_sample = median(assessment_df$num_var) # my solution (don't use this because using the $ there i
median_samp = median(pull(assessment_df, num_var)) # actual solution (tidyverse solution)

median_sample == median_samp
```

```
## [1] TRUE
```

```
median_sample
```

```
## [1] 0.9446332
```

The median value is 0.94

why is my in-line different from the code chunk?? (probably because i didn't set seed? yeah that's why)

Calculating the median value inline 0.94

Section 6: formatting

Text formatting

italic or *italic* **bold** or **bold** code superscript² and subscript₂

superscript³ and subscript₃

Headings

1st Level Header

2nd Level Header

3rd Level Header

Lists

- Bulleted list item 1
 - Item 2
 - Item 2a
 - Item 2b
1. Numbered list item 1
 2. Item 2. The numbers are incremented automatically in the output.

Tables

First Header	Second Header
Content Cell	Content Cell
Content Cell	Content Cell

Section 7: Learning Assessment 3

After the previous code chunk, write a bullet list given the mean, median, and standard deviation of the original random sample.

- The median value is 0.94
- The mean value is 1
- The standard deviation is 0.99